Guiding Mixture-of-Experts Reasoning via Routing Intervention

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Current experimental scope. All results available at the time of writing use `allenai/OLMoE-1B-7B-0924-Instruct` (16 MoE layers, 64 experts, top-$k = 8$). The current successful runs are GSM8K (1,319 generated traces), GSM8K with a self-check prompt (1,319), MATH (1,500), MATH with a self-check prompt (1,500), and ProcessBench (400 given-solution traces). All five have complete router and hidden-state tensors and outputs for Experiments 1–5. The current PRM800K launch is not a completed experiment: its Experiment 1 JSONL is empty, and downstream stages produced zero-input placeholder files. Older, corrected-label PRM800K event and full-trace probe outputs over 10,833 traces are retained as explicitly marked legacy evidence. Unless stated otherwise, event windows contain five tokens on each side of the event.

Implemented analysis pipeline. The repository implements trace generation or ingestion (Experiment 1), token-level router and optional hidden-state extraction (Experiment 2), event-centered routing with matched within-trace controls, geometry–routing correlation (Experiment 3), prospective prefix probes (Experiment 4), and expert phase/co-activation/transition analysis (Experiment 5). Offline tools also support high-precision relabelling, rebuilding multi-dataset taxonomy summaries, full-trace layerwise linear probes over router, hidden, or combined features, and a trace-length/structure baseline. The full pipeline runs Experiments 1–5 and extracts hidden states by default; the latter and Experiment 4 can be disabled together when storage is constrained.

Prospective probe design and status. Experiment 4 pools only the first 25%, 50%, or 75% of each trace and trains separate trace-grouped logistic probes from structure-only, router-only, hidden-only, and combined hidden/router features. It evaluates final incorrectness and whether a gold first error remains beyond the observed prefix, excluding traces whose error has already occurred for the latter target. Outputs include out-of-fold AUROC, balanced accuracy, F1, and trace-bootstrap confidence intervals. This pipeline is implemented and has completed for all five current conditions. Final-incorrectness probes are available everywhere. Future-first-error probes are available only for ProcessBench because the generated datasets do not have gold error-step annotations.

Important provenance note. The event-label heuristic was made substantially more conservative after the first experimental run. Taxonomy, event-routing, and expert-event outputs for the five current conditions were regenerated with the corrected labels, and their JSONL files pass line-by-line JSON validation. The separate PRM800K relabelled files and full-trace probe outputs predate the current pipeline run and are never pooled with current estimates. SLURM's printed "Pipeline complete" banner is not treated as evidence of success:

Author's Contact Information:

DOI:

Table 1. High-precision trace taxonomy. Percentages are of all traces. SC denotes self-correction; the PRM800K row is a corrected legacy output rather than part of the current five-condition run.

Dataset	N	Accuracy	Backtracking	Contradiction	SC	Final-answer reversal
GSM8K	1,319	37.38%	1 (0.076%)	1 (0.076%)	1 (0.076%)	0 (0.000%)
GSM8K + self-check	1,319	36.01%	9 (0.682%)	3 (0.227%)	7 (0.531%)	0 (0.000%)
MATH	1,500	13.47%	0 (0.000%)	15 (1.000%)	0 (0.000%)	28 (1.867%)
MATH + self-check	1,500	11.80%	9 (0.600%)	19 (1.267%)	9 (0.600%)	40 (2.667%)
ProcessBench	400	48.25%	3 (0.750%)	0 (0.000%)	3 (0.750%)	1 (0.250%)
PRM800K (legacy)	10,833	60.17%	2 (0.018%)	87 (0.803%)	1 (0.009%)	10 (0.092%)

both PRM800K logs contain explicit zero-input errors even though the shell script continued to that banner. Historical logs also include a CUDA out-of-memory failure and several OLMoE/Qwen generations whose logs end mid-progress; no Qwen result artifacts are present, so these are excluded rather than counted as robustness runs. Repeated cgroup swap-limit warnings also appear in the successful jobs and are treated as environmental warnings, not failures.

6 Results

6.1 Reasoning outcomes and event taxonomy

Table 1 is recomputed directly from the current trace JSONL files; the PRM800K row alone comes from the separately retained corrected-label output. Accuracy is final-answer accuracy for generated traces and the stored outcome label for given solutions. Explicit revision language remains rare, but self-check prompting increases its incidence: backtracking rises from 1 to 9 traces on GSM8K and from 0 to 9 on MATH. This does not translate into successful repair. All seven GSM8K-self-check and all nine MATH-self-check traces labelled as self-corrections end incorrect.

Because baseline and self-check generations use the same problem IDs, we also compare correctness within problem. On GSM8K, self-checking changes 168 baseline-correct answers to incorrect and 150 baseline-incorrect answers to correct (exact McNemar $p = 0.340$). On the 1,499 MATH problems with determined correctness in both conditions, the corresponding counts are 107 and 82 ($p = 0.081$). Thus neither dataset provides evidence that the self-check prompt improves accuracy; its observed point estimate is lower by 1.37 and 1.67 percentage points, respectively. The increased event count should therefore be interpreted as elicited revision language, not effective self-correction.

The earlier relabelling audit in Table 2 remains an important warning: the original phrase matcher had many false positives. For example, GSM8K backtracking labels fell from 10 to 1, ProcessBench from 14 to 3, and PRM800K from 112 to 2. Claims based on the old labels are discarded. Even after deliberate self-check elicitation, current trace counts of 7–9 self-corrections are too small for broad claims.

6.2 Event-centered routing

The most reliable event-centered result currently available is the ProcessBench analysis at the gold first-error step ($n = 207$ traces with an error). Table 3 reports the raw trajectory. All changes are small. The matched within-trace control analysis in Table 4 confirms the null result: every 95% bootstrap confidence interval includes zero. Thus, with the present layer-averaged metrics, there is no evidence of a discontinuous routing change at the first erroneous step.

Table 2. Number of traces carrying each label before and after the high-precision relabelling.

Dataset	Version	Backtrack	Contradiction	Self-correction
GSM8K	old	10	5	9
	current	1	1	1
PRM800K	old	112	145	61
	current	2	87	1
ProcessBench	old	14	2	11
	current	3	0	3

Table 3. ProcessBench routing around the gold first-error step. Values are averaged across traces and layers.

Metric	Before	At first error	After
Routing entropy	3.80831	3.80624	3.80520
Expert-switch rate	0.60402	0.59066	0.58478
Top-k overlap	0.27355	0.28462	0.28982
Router margin	0.01986	0.01940	0.01953

Table 4. Matched-control effect at the ProcessBench first-error step. The effect is event change minus change at a matched normal position in the same trace; $n = 207$, 2,000 bootstrap samples.

Metric	Mean matched-control effect	95% bootstrap CI
Routing entropy	0.00378	$[-0.00878,\ 0.01587]$
Expert-switch rate	-0.00759	$[-0.03295,\ 0.01876]$
Top-k overlap	0.00712	$[-0.01553,\ 0.02996]$
Router margin	-0.000318	$[-0.001506,\ 0.000844]$

Plain GSM8K still has only two event-eligible traces, so its event effects are case studies. Self-checking increases GSM8K coverage to ten eligible traces, but every bootstrap interval for backtracking, self-correction, and the first detected event contains zero. Table 5 summarizes the better-powered generated-data comparisons. Two small MATH entropy effects are resolved: contradiction in the baseline condition and the first detected event under self-checking. In MATH self-check backtracking/ self-correction spans, the router margin falls. The latter labels identify the same nine traces and nearly the same spans, so they are not independent replications. None of these analyses corrects for the many event–metric comparisons, and trace counts are only 9–24; they are exploratory rather than evidence of a universal event signature.

The retained corrected PRM800K output contains 4,315 gold first-error traces, 87 traces with explicit contradictions, and 89 first heuristic reasoning events. Table 6 reports its earlier matched analysis. Contradiction and first-event intervals include zero. At the gold first error, switching falls, overlap rises, and margin rises by small amounts. It is a useful counterexample to a universal disruption hypothesis, but it is legacy evidence: the current PRM800K pipeline did not regenerate it.

Legacy PRM800K backtracking ($n = 2$) and self-correction ($n = 1$) remain case studies and are omitted from the inferential table.

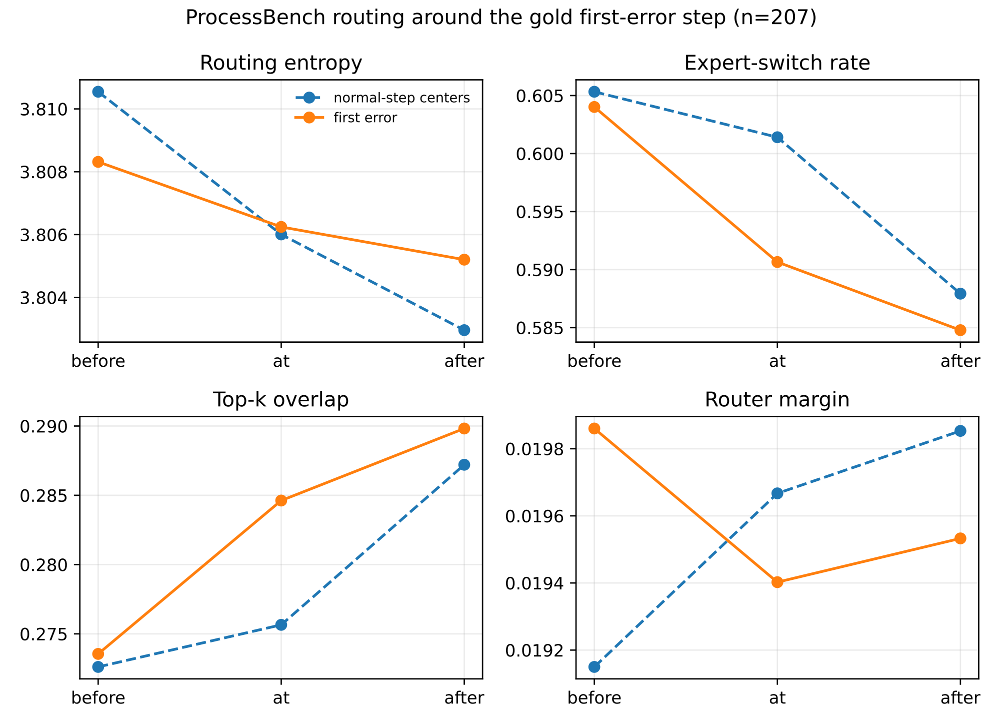

Fig. 1. Before/at/after trajectories for ProcessBench gold first errors and normal-step centers. The visual differences are descriptive; the controlled effects in Table 4 are not statistically resolved.

Table 5. Current generated-data matched-control effects. Values are event change minus change at a matched normal position; intervals are trace-level 95% bootstrap intervals with 2,000 replicates.

Condition	Event	n	Δ entropy	Δ switch	Δ overlap	Δ margin
GSM8K + self-check	Backtracking	9	−0.00670 [−0.04040, 0.03064]	0.01237 [−0.06946, 0.09246]	−0.00226 [−0.07058, 0.06825]	−0.000833 [−0.004740, 0.003242]
GSM8K + self-check	Self-correction	7	−0.01062 [−0.05124, 0.03238]	0.03934 [−0.02679, 0.11244]	−0.02583 [−0.10064, 0.03729]	−0.000534 [−0.005338, 0.004480]
MATH	Contradiction	15	0.04381 [0.01039, 0.07702]	0.02031 [−0.06303, 0.10522]	−0.01834 [−0.09607, 0.06839]	−0.000884 [−0.005099, 0.003685]
MATH + self-check	First detected event	24	0.04130 [0.00639, 0.07399]	0.03743 [−0.03776, 0.10645]	−0.02995 [−0.09325, 0.03043]	−0.002898 [−0.005988, 0.000365]
MATH + self-check	Backtracking	9	0.03671 [−0.02635, 0.09093]	0.04991 [−0.04862, 0.13413]	−0.04524 [−0.11074, 0.03742]	−0.003962 [−0.007604, −0.000631]

Table 6. Legacy corrected PRM800K matched-control effects at the event. Intervals are trace-level 95% bootstrap intervals with 1,000 replicates.

Event	n traces	Δ entropy	Δ switch	Δ overlap	Δ margin
Contradiction	87	−0.00210 [−0.02381, 0.01722]	0.00576 [−0.03197, 0.04148]	−0.00324 [−0.03431, 0.02971]	0.000146 [−0.001789, 0.002247]
First reasoning event	89	−0.00491 [−0.02597, 0.01449]	0.00888 [−0.02478, 0.04288]	−0.00637 [−0.03866, 0.02663]	0.000511 [−0.001572, 0.002594]
Gold first error	4,315	−0.00187 [−0.00485, 0.00107]	−0.02120 [−0.02687, −0.01577]	0.01980 [0.01457, 0.02515]	0.000474 [0.000158, 0.000801]

6.3 Hidden-state geometry and routing

For each layer, we compute the pairwise cosine-similarity matrix of sampled token hidden states and the cosine-similarity matrix of their softmax router distributions. Their upper triangles are correlated; the overall significance

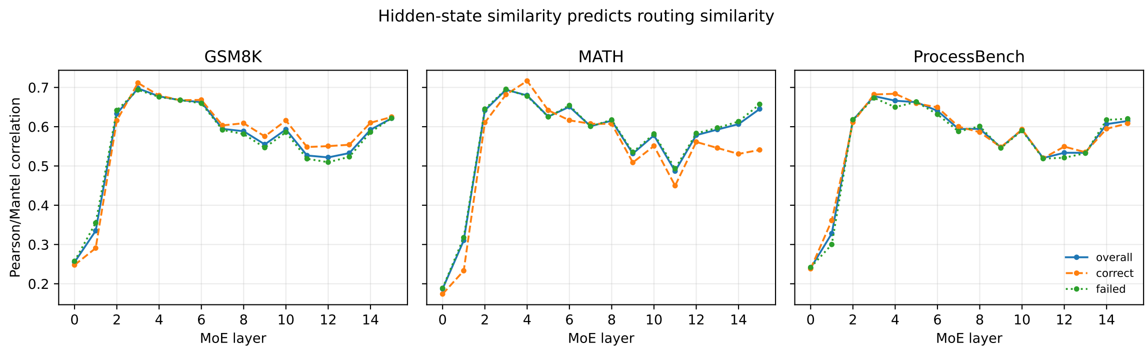

Fig. 2. Layerwise correlation between hidden-state similarity and routing- distribution similarity for three representative current conditions.

Table 7. Summary of geometry–routing correlations across 16 layers.

Condition	Overall mean	Overall range	Peak layer	Correct mean	Failed mean
GSM8K	0.5659	0.2550–0.6977	3	0.5732	0.5635
GSM8K + self-check	0.5644	0.2529–0.6936	3	0.5830	0.5597
MATH	0.5641	0.1870–0.6940	3	0.5360	0.5676
MATH + self-check	0.5663	0.2194–0.6809	4	0.5655	0.5684
ProcessBench	0.5605	0.2399–0.6779	3	0.5637	0.5570

test is a 1,000-permutation Mantel test. This is a geometry– routing correlation analysis, not t-SNE and not a linear-separability probe. The exposed hidden state is the input to the transformer layer and is the closest available proxy for, but not exactly, the post-attention state seen by that layer's router.

Figure 2 and Table 7 show the most replicated result in the study. Mean overall correlation lies in the narrow range 0.5605–0.5663 across all five current conditions. It rises sharply from layer 0 and peaks in layer 3 or 4 before declining moderately. Every layer in every condition reaches the permutation-resolution floor $p = 1/1001 \approx 0.000999$.

Correct-versus-failed differences are inconsistent. Correct traces have the higher mean on GSM8K, GSM8K self-check, and ProcessBench; failed traces have the higher mean on MATH and (slightly) MATH self-check. No trace-bootstrap interval is available for these contrasts. The overall analysis uniformly samples 1,000 tokens per condition and the Mantel permutation treats sampled tokens as exchangeable; it does not provide trace-level uncertainty or prove that representations cause routing. The defensible conclusion is narrower: hidden-state and router-distribution geometries are strongly aligned and the layerwise profile replicates across datasets and prompts, while outcome modulation does not.

6.4 Prospective prefix probes

Experiment 4 now supplies the prospective test that was previously missing. Table 8 reports early (25%) and late (75%) prefixes; the omitted 50% results lie between them except where noted below. On GSM8K and both MATH conditions, a three-feature length/step/character baseline is as good as or better than router, hidden, and

Table 8. Prospective final-incorrectness AUROC from prefix-only features. Representation entries select the best of 16 layers (shown in parentheses), so their comparison is exploratory and selection-optimistic. “Positive” is the number of finally incorrect traces.

Condition	Prefix	N	Positive	Structure	Router	Hidden	Combined
GSM8K	25%	1,319	826	0.6819	0.6512 (L10)	0.6559 (L12)	0.6514 (L10)
	75%	1,319	826	0.6819	0.6887 (L13)	0.6839 (L12)	0.6830 (L8)
GSM8K + self-check	25%	1,319	844	0.6960	0.6732 (L7)	0.6880 (L9)	0.6874 (L10)
	75%	1,319	844	0.6959	0.6843 (L9)	0.7070 (L10)	0.7105 (L10)
MATH	25%	1,499	1,297	0.7628	0.7144 (L12)	0.7189 (L14)	0.7252 (L14)
	75%	1,499	1,297	0.7628	0.7319 (L10)	0.7445 (L12)	0.7370 (L6)
MATH + self-check	25%	1,500	1,323	0.7736	0.7118 (L10)	0.7216 (L14)	0.7112 (L14)
	75%	1,500	1,323	0.7735	0.7446 (L10)	0.7420 (L15)	0.7402 (L15)
ProcessBench	25%	400	207	0.6192	0.6267 (L9)	0.6171 (L7)	0.6171 (L6)
	75%	400	207	0.6184	0.6685 (L4)	0.6931 (L11)	0.7022 (L11)

combined representations at most prefixes. MATH is especially clear: structure AUROC is approximately 0.763–0.774 while the best representation probe reaches at most 0.745. This makes trace structure a major confound, not merely a weak baseline.

ProcessBench is the exception for final incorrectness: combined AUROC rises from 0.617 at 25% to 0.702 at 75%, versus a 0.618 structure baseline. The signal therefore emerges after much of the solution is visible, which is useful for monitoring but weaker evidence of advance prediction.

The stricter future-first-error target is available only on ProcessBench. At 25%, combined features reach AUROC 0.674 (95% bootstrap CI $[0.612, 0.733]$) versus 0.625 for structure. This advantage disappears at 50%, where structure is 0.640 and combined features are 0.563. At 75%, only 13 future errors remain; the best router AUROC is 0.682 with the very wide interval $[0.535, 0.832]$. Fractions condition on the error not having occurred yet, so their populations and class prevalence differ and should not be read as a learning curve. Together with best-layer selection and the absence of paired model-comparison tests, these results do not establish a robust advance failure detector.

6.5 Expert usage by reasoning phase

Experiment 5 measures each expert’s activation frequency and weight mass by phase, before/after event usage, co-activation, and top-1 transition matrices. The outputs in Table 9 and Figure 3 use current labels. Each cell reports the mean absolute activation-frequency difference over all 16×64 layer–expert cells relative to normal tokens, followed by the number of phase tokens. Token counts are not numbers of independent traces; in particular, explicit event phases remain supported by only 1–9 traces per condition.

The reliable ProcessBench first-error phase has only a 0.0104 mean absolute shift from normal expert frequencies. Comparing the five tokens after each gold first error with the five before it gives a mean absolute layer–expert change of 0.0139 over 207 events; the largest cell change is 0.0879. For the self-check conditions, five-token before/after changes at self-corrections are 0.0521 over ten GSM8K event occurrences and 0.0436 over ten MATH occurrences, but every associated trace ends incorrect and no matched-control interval is available. These are state changes around failed revision attempts, not evidence of recovery experts.

Table 10 adds the marginal-frequency baseline missing from the preliminary analysis. For each layer, the baseline is the self-transition probability under independent source and destination draws with the observed

Table 9. Expert-activation shifts relative to normal tokens. Parentheses give phase-token counts; a dash indicates no tokens for that phase.

Dataset	Backtrack	Contradiction	Self-correction	First error	Final answer
GSM8K	0.0608 (61)	0.0804 (82)	0.0608 (61)	–	0.0136 (139,075)
GSM8K + self-check	0.0266 (2,512)	0.0390 (670)	0.0303 (1,997)	–	0.0129 (70,550)
MATH	–	0.0224 (1,677)	–	–	0.0160 (167,642)
MATH + self-check	0.0175 (2,345)	0.0154 (4,085)	0.0175 (2,345)	–	0.0143 (119,763)
ProcessBench	0.0592 (162)	–	0.0592 (162)	0.0104 (14,539)	0.0185 (13,210)

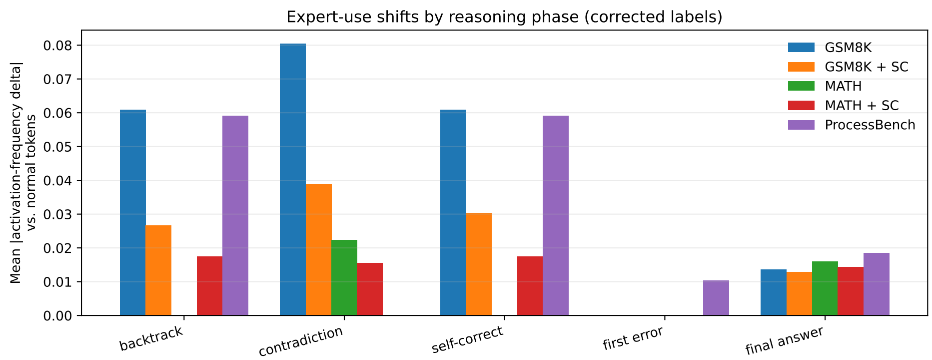

Fig. 3. Mean absolute expert-activation-frequency shift by reasoning phase. Explicit backtracking/self-correction bars have very small trace counts even when they contain many tokens.

Table 10. Global top-1 expert transition persistence and a marginal-frequency independence baseline.

Condition	Transitions	Self-transition	Independence
GSM8K	5,398,816	19.79%	9.90%
GSM8K + self-check	5,456,304	19.30%	9.82%
MATH	10,396,432	29.86%	20.58%
MATH + self-check	10,390,832	29.49%	20.14%
ProcessBench	1,681,792	20.54%	10.19%

marginals, then pooled by transition count. Observed persistence is 9–10 percentage points above this baseline in every current condition. The effect is descriptive—there is no trace-level uncertainty and token identity/content are uncontrolled—but it does show that the roughly 19–30% self-transition shares are not explained solely by expert imbalance.

6.6 Legacy full-trace linear routing probes

Before the current prospective run, we trained a separate linear logistic probe at each layer using one pooled router-feature vector per trace. The 197 features comprise router-probability means and standard deviations, top-1 expert frequencies, entropy and margin statistics, and top-1 switch rate. Five-fold stratified cross-validation

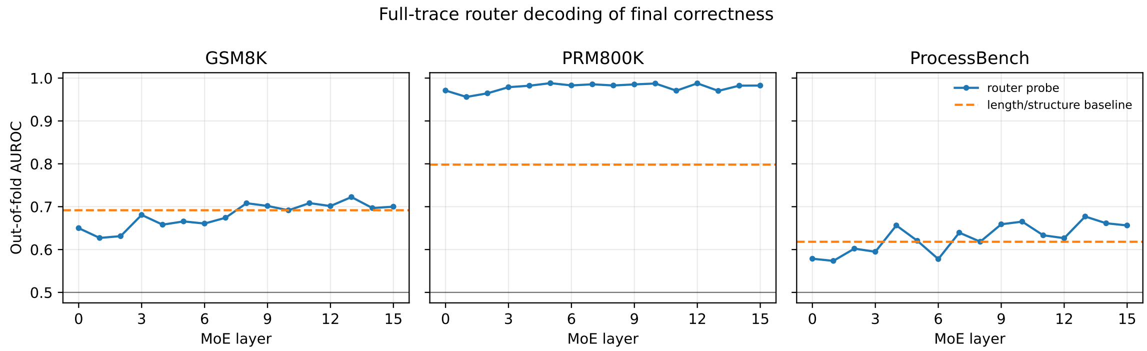

Fig. 4. Out-of-fold correctness AUROC by layer. Dashed lines use only log token count, log step count, and log character count.

Table 11. Legacy best-layer full-trace router probes. The selected-layer comparison is exploratory and is not corrected for selecting the best of 16 layers.

Dataset	Target	*N*	Positive	Best layer	Router AUROC [95% CI]	Balanced acc.	Structure AUROC
GSM8K	Correctness	1,319	486	13	0.7225 [0.6944, 0.7505]	0.6655	0.6917
PRM800K	Correctness	10,833	6,518	5	0.9882 [0.9858, 0.9905]	0.9760	0.7981
PRM800K	Gold first error	10,833	4,315	5	0.9882 [0.9857, 0.9904]	0.9760	0.7981
PRM800K	Contradiction	10,833	87	9	0.9082 [0.8702, 0.9385]	0.7489	0.7573
ProcessBench	Correctness	400	193	13	0.6771 [0.6244, 0.7272]	0.6356	0.6181
ProcessBench	Gold first error	400	207	13	0.6771 [0.6240, 0.7303]	0.6356	0.6181

produces out-of-fold predictions; 95% intervals bootstrap traces. Since each trace contributes one sample, token leakage across folds is impossible. The probe pools the *complete* trace, so this is a decoding/separability test, not online prediction of a future error. These files are retained because they include the only nonempty PRM800K probe, but the GSM8K generations and labels are not identical to the current run; they are not compared numerically with Table 8.

Routing improves over the simple structure baseline by 0.031 AUROC on GSM8K, 0.190 on PRM800K correctness/first-error, 0.151 on PRM800K contradiction, and 0.059 on ProcessBench. PRM800K correctness and first-error scores are the same because, in these traces, presence of a gold first error is the complement of the final correctness label. The contradiction AUROC is high despite only 87 positives, but F1 at the AUROC-selected layer is only 0.159 under the natural class imbalance. Backtracking and self-correction probes were correctly skipped: their positive counts after relabelling are 1/1 on GSM8K, 2/1 on PRM800K, and 3/3 on ProcessBench.

The near-perfect PRM800K correctness AUROC (0.988) should be treated with particular caution. The probe sees the complete provided solution, the first- error target is exactly complementary to correctness in this output, and source/format artifacts can be encoded in pooled routing statistics. The current zero-input PRM800K failure prevents a prefix-only hidden/router check, so this result establishes legacy within-file decodability only.

7 Analysis

7.1 Conclusions supported so far

(1) Hidden-state geometry strongly predicts router-distribution geometry at every layer in all five current conditions. Overall mean correlation is 0.5605–0.5663, peaks in layer 3 or 4, and is nearly invariant to self-check prompting. Correct–failed differences reverse direction across conditions.
(2) At 207 gold ProcessBench first errors, entropy, switching, overlap, and margin do not exhibit a statistically resolved discontinuity relative to a matched position in the same trace. Simple layer-averaged routing metrics are therefore not sufficient first-error markers in the current analysis.
(3) Self-check prompting increases explicit revision markers but does not improve final accuracy on paired GSM8K or MATH problems. Every trace labelled as a self-correction in the two self-check conditions ends incorrect.
(4) MATH exhibits a small entropy increase at detected reasoning events, and MATH self-check backtracking spans exhibit a small margin decrease. These effects use only 9–24 traces, are uncorrected for multiple comparisons, and do not replicate as a universal multi-metric disruption.
(5) Prospective final-incorrectness probes are usually matched or beaten by the structure baseline on GSM8K and MATH. ProcessBench combined features reach AUROC 0.702 at 75%, but its 25% future-first-error result is only 0.674 and the signal is not stable across later conditional cohorts. A robust advance detector has not been established.
(6) Top-1 routes persist more than a marginal-frequency independence baseline in every current condition, but event-specific expert shifts remain descriptive and explicit-event trace counts remain small.
(7) The current PRM800K run failed with zero input despite completion banners. Its earlier first-error stabilization and near-perfect full-trace decoding results are legacy evidence and require a clean prospective rerun.

7.2 What has not been established

The present experiments do not identify "math experts" or "failure experts", do not show that routing changes cause reasoning errors, and do not show that backtracking forms a stable, linearly separable hidden-state class. Hidden and combined prefix probes have now been run, but their advantage over structure is inconsistent and best-layer comparisons are selection-optimistic. No t-SNE/UMAP visualization or routing intervention has been run. The geometry significance test does not supply trace-level uncertainty, event analyses are not multiplicity-corrected, and the expert transition baseline does not control token content. Correlation and linear decodability should not be read as causal evidence.

8 Discussion

The strongest current story remains geometry-first: routing follows the evolving representation space, with a consistent layerwise signature across five dataset and prompt conditions. The anticipated universal sharp disruption at errors is not supported. ProcessBench is null at the gold first error, MATH has a selective entropy increase at heuristically detected events, and legacy PRM800K shows a small stabilization. Direction and metric therefore depend on the dataset and event definition.

The prospective runs also change the interpretation of decodability. Much of final-incorrectness prediction on generated math traces is already present in prefix length and structure; representation features add most clearly only late in ProcessBench. Future work should use paired model comparisons, prespecified layers, trace-level uncertainty, and targets defined strictly after the observed prefix. A clean PRM800K rerun is necessary before treating its unusually high legacy decoding score as a model-level phenomenon.

The event-label audit also changes the experimental priority. Explicit lexical markers provide high precision but extremely low recall. Robust backtracking conclusions will require either deliberately elicited self-checking traces, manual/LLM-assisted span annotation with validation, or a benchmark with revision spans. Until then, the gold ProcessBench error analysis and outcome-based correct/failed comparisons are the safest results.

9 Conclusion

OLMoE routing is strongly and reproducibly aligned with hidden-state geometry, but aggregate metrics do not provide a dataset-invariant error signature. Self-check prompting elicits more revision language without improving paired accuracy, and prospective representation probes seldom beat simple structure baselines on generated math. ProcessBench shows a moderate late-prefix signal but no resolved discontinuity at its gold first error. Expert routes are more persistent than their marginal frequencies predict, yet event- specific expert claims remain underpowered. The next priorities are a clean PRM800K rerun, prespecified/paired prospective evaluation, stronger event supervision, and only then causal routing intervention.

A Reproducibility Checklist

Table 12. Living inventory of completed and pending work.

Item	Status	Note / next action
Trace generation / ingestion	Partial	Current: GSM8K and GSM8K self-check 1,319 each; MATH and MATH self-check 1,500 each; ProcessBench 400. PRM800K is zero-byte and must be rerun.
Router/hidden extraction	Done for five conditions	All five non-PRM JSONL files validate and have complete router/hidden tensors; PRM800K current output is empty.
High-precision labels	Done/current	Current five-condition outputs use the conservative classifier; legacy PRM800K has a separate corrected-label file.
Taxonomy summary file	Done/current	`results/exp1/summary.json` contains the five current conditions; corrected legacy PRM800K remains in `summary_relabelled.json`.
ProcessBench event routing	Done/current	Gold first error has matched controls and 2,000-sample bootstrap CIs.
Generated-data event routing	Done but underpowered	Plain/self-check GSM8K and MATH are complete; event-eligible trace counts range from 2 to 24.
PRM800K event routing	Legacy only	Corrected earlier output covers 4,371 event-eligible traces; current zero-input run did not reproduce it.
Geometry–routing correlation	Done for five conditions	Overall/correct/failed and current-label backtracking outputs exist for every non-PRM condition.
Expert-event analysis	Done for five conditions	Current JSON/NPZ outputs exist; explicit event trace counts remain underpowered.
t-SNE/UMAP	Not done	Optional visualization only; stratify by layer and trace split if added.
Legacy full-trace probes	Done/legacy	Trace-grouped router and structure probes exist for GSM8K, PRM800K, and ProcessBench but are not current-run prospective results.
Hidden/combined prefix probes	Done for five conditions	Experiment 4 compares structure, router, hidden, and combined 25/50/75% prefix features.
Prospective failure prediction	Partially established	Final incorrectness is evaluated everywhere; future first error is evaluable only on ProcessBench and is not stable across conditional cohorts.
Structural/MFA analysis	Not done	Add trajectory length/curvature, reasoning structures, and optional local-geometry regions after core probes.
Routing intervention	Not done	Title-level intervention claim is not yet supported; run only after stable predictive signals are identified.
Second MoE / robustness	Not done	Run a Qwen MoE or another routing design once the corrected pipeline is frozen.
Pipeline status handling	Defective	The shell script prints completion after zero-input stage errors; add nonzero exits, output validation, and atomic JSONL writes.
Figures	Current after regeneration	Geometry, ProcessBench first-error, expert-phase, and legacy router-probe plots are generated by `paper_figures/generate_figures.py`.